Problem Set 3

Data Visualisation for Social Scientists

Due: February 18, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Wednesday February 18, 2026. No late assignments will be accepted.

Canadian Election Study

The data for this problem set come from the Canadian Election Study (CES) in 2015. The main purpose of the study is to give a comprehensive picture of the Canadian election: why people vote as they do, what changes during campaigns and across elections, and how Canadian voting compares with that in other democracies.

Data Manipulation

1. Load the CES .csv file from GitHub into your global environment. Filter respondents to only include "high quality" participants:

```
ces2015 <- ces2015 |> filter(discard == "Good quality")
```

2. Filter the dataset to those participants that answered the question about voting for the past election using `p_voted`. Consider respondents who gave a "Yes" answer as having voted, while "No" as not having voted. Treat "Don't know" and "Refused" as missing.

3. Create an age variable and group into categories (e.g., <30, 30-44, 45-64, 65+). Year of birth is in age (four-digit year).

Answer 1

The dataset is loaded in the global environment and the desired variable, **discard**, is subselected, only keeping *Good Quality* observations.

```
ces2015_raw <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/DataViz_
    2026/refs/heads/main/datasets/CES2015.csv")

ces2015 <- ces2015_raw|>
  filter(discard == "Good quality")
```

Answer 2

The variable **p_voted** is mutated as to recode the replies *Don't Know* and *Refused* into NAs. The *Yes* and *No* replies were kept as such.

```
ces2015 <- ces2015 |>
  mutate(
    p_voted = case_when(
      p_voted == "Yes" ~ "Yes",
      p_voted == "No" ~ "No",
      p_voted %in% c("Don't know", "Refused") ~ NA_character_
    ))
```

Answer 3

In order to create a categorical variable the age of respondents is calculated by subtracting the **age** value from 2015, when the dataset was collected.

```
ces2015 <- ces2015 |>
  mutate( age = as.numeric(age),
          age_years = 2015 - age )|>
  #removing wrongly coded observation
  filter(age_years != 1015)
```

Afterwards the **age_cat** variable is created by binning the **age_years** variable, four categories are obtained.

```
ces2015$age_cat <- cut(
  ces2015$age_years,
  breaks = c(-1, 29, 44, 64, 115),
```

```
  labels = c("<30", "30-44", "45-64", "65+"),
  right = TRUE)
```

Data Visualization

1. Plot turnout rate by age group.

2. Create a density plot of ideology by party, restricting your sample to respondents with non-missing left–right self-placement (0–10 scale) and those that intended to vote for a main party (e.g., Liberal, Conservative, NDP, Bloc in Quebec, and Green).

3. Produce histogram counts of turnout by income (`income_full`), faceted by province.

4. Create your own reusable custom theme. Apply your theme to one of the previous plots and add:

   (a) An improved title summarizing the main substantive takeaway.

   (b) A more informative subtitle describing the sample and variables.

   (c) A caption noting data source, weighting, and key coding decisions.

   (d) At least one direct annotation using `ggrepel` that calls out a key pattern.

Answer 1

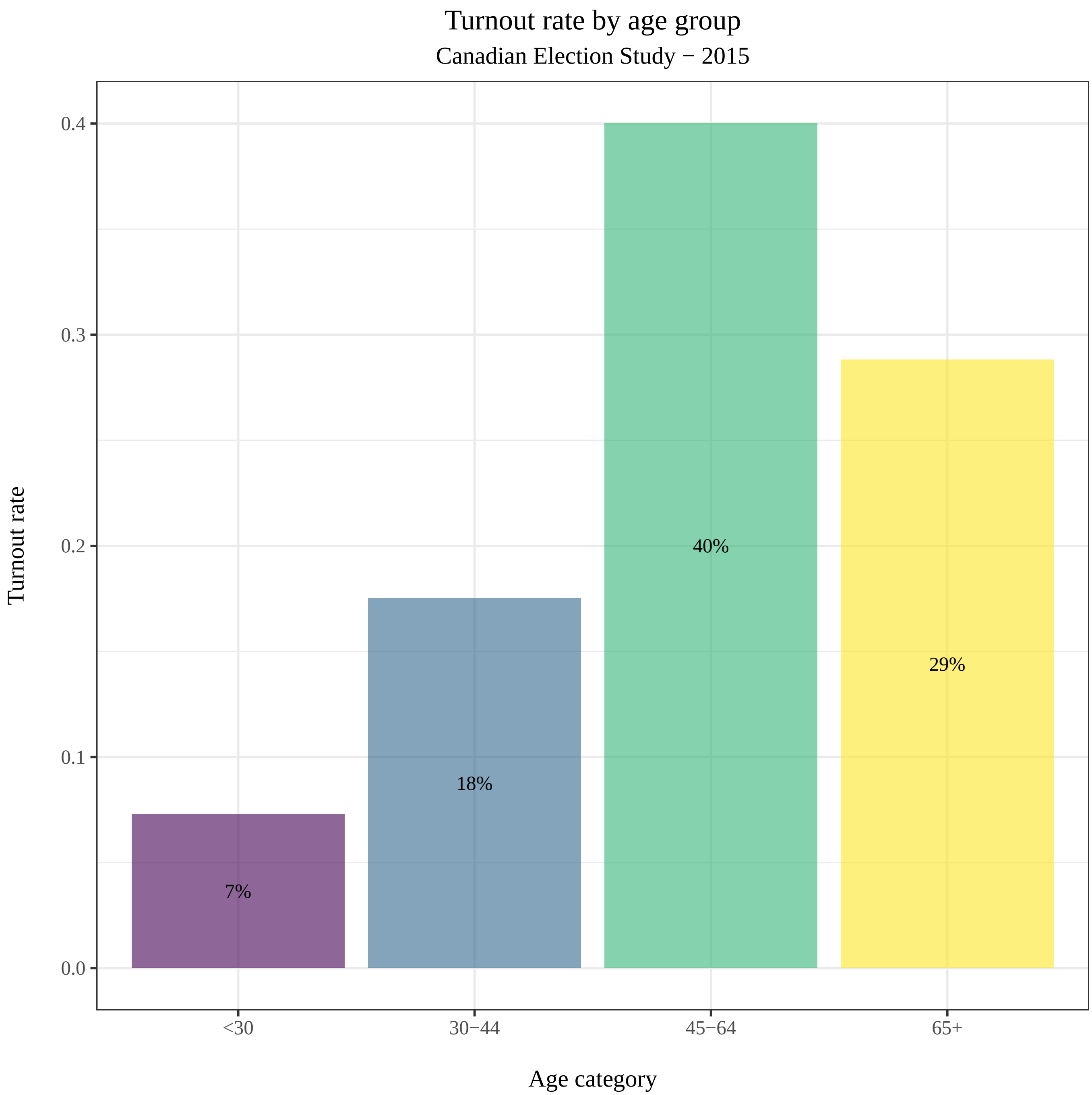
To calculate the turnout rate for each age group, we first determine the total number of valid voters, *total_valid*, defined as those who answered either *Yes* or *No* to the *p_voted* variable. Next, for each age category, the **turnout_rate** is calculated by dividing the number of respondents who answered *Yes* by *total_valid*.

```
total_valid <- ces2015 |>
  filter(p_voted %in% c("Yes", "No")) |>
  nrow()

ces2015_viz1 <- ces2015 |>
  filter(p_voted %in% c("Yes", "No")) |>
  group_by(age_cat) |>
  summarise(
    yes_count = sum(p_voted == "Yes", na.rm = TRUE),
    turnout_rate = yes_count / total_valid
  )
```

Figure 1 shows the turnout rate for each age group relative to the total number of valid voters. The 45–64-year-old group has the highest turnout rate at 40%, followed by the 65+ age group. Turnout is noticeably lower among younger voters, particularly those aged 18–24, indicating lower electoral engagement. Overall, turnout tends to increase with age, suggesting that middle-aged and older citizens are more likely to participate in elections.

```
ggplot(data = ces2015_viz1, aes(x = age_cat, y = turnout_rate, fill = age_cat)
  ) +
  geom_col(alpha = 0.6) +
  geom_text(aes(label = scales::percent(turnout_rate, accuracy = 1)),
            position = position_stack(vjust = 0.5),
            size = 3) +
  scale_fill_viridis_d(name = "age_cat") +
  labs(title = "Turnout rate by age group",
       subtitle = "Canadian Election Study - 2015",
       x = "\nAge category",
       y = "Turnout rate\n") +
  theme_bw() +
  theme(
    text = element_text(family = "Times"),
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    legend.position = "none"
  )
```

Figure 1:



Answer 2

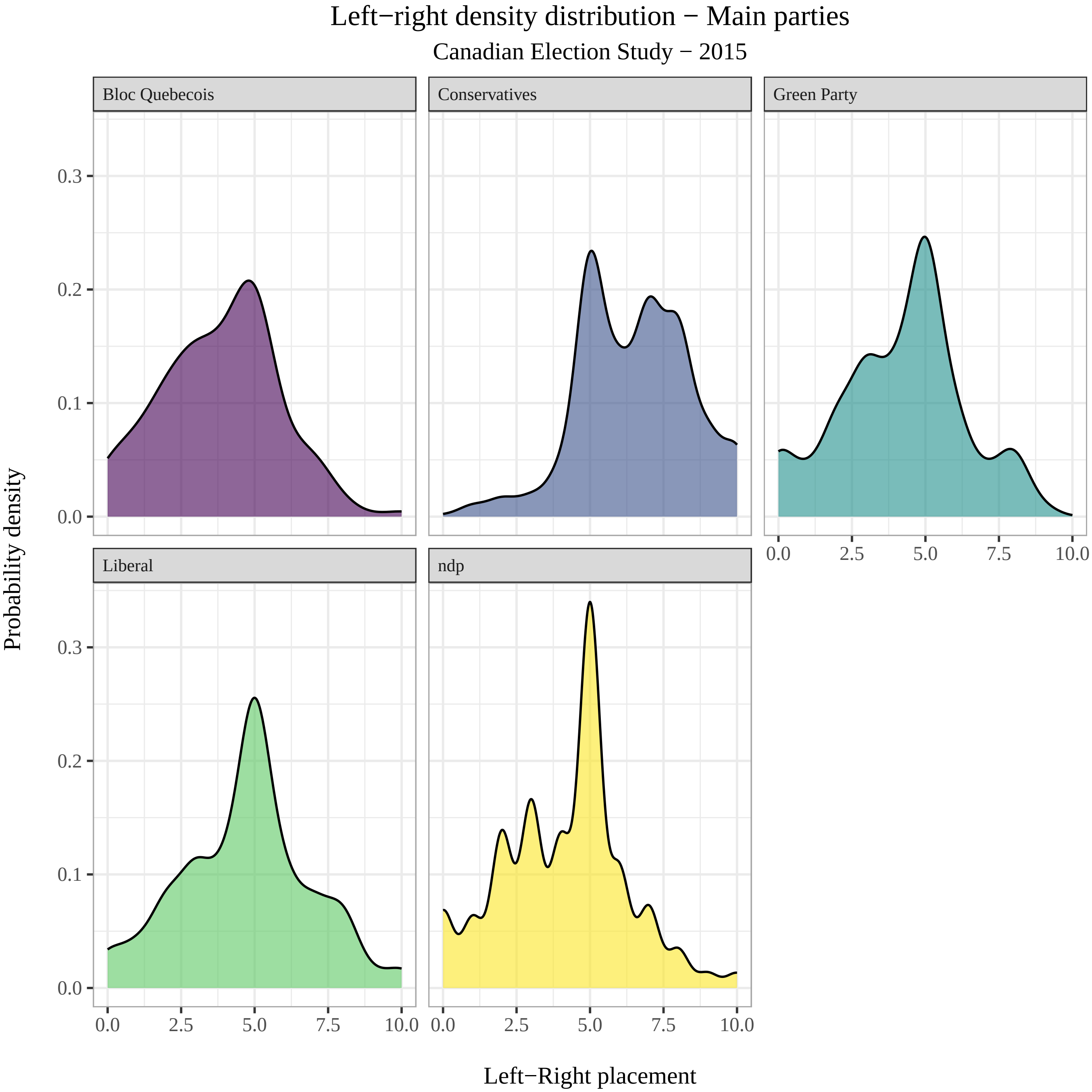
To create the second visualization both the **p_selfplace** and the **vote_for** are filtered as to only keep those who responded in the first, and the desired main party voters in the second.

```
ces2015_viz2 <- ces2015 |>
  mutate(p_selfplace = as.numeric(p_selfplace))|>
  filter(vote_for %in% c(
    "Liberal",
    "Conservatives",
    "ndp",
    "Bloc Quebecois",
    "Green Party"),
  p_selfplace %in% 0:10)
```

Figure 2 shows the density plot of left–right ideology placement for each of the five main Canadian parties. The *Conservatives* and *Bloc Québécois* are the most strongly skewed toward the extremes, to the right and left respectively. The other three parties are more centrist, exhibiting a higher central peak in their distributions.

```
ggplot(data = ces2015_viz2, aes(x = p_selfplace, fill = vote_for))+
  geom_density(alpha = 0.6) +
  facet_wrap(vars(vote_for)) +
  scale_fill_viridis_d(name = "vote_for")+
labs(title = "Left-right density distribution - Main parties",
     subtitle = "Canadian Election Study - 2015",
     x = "\nLeft-Right placement",
     y = "Probability density\n") +
  theme_bw() +
  theme(
    text = element_text(family = "Times"),
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    strip.text = element_text(size = 8 , hjust = 0),
    panel.border = element_rect(color = "darkgrey"),
    legend.position = "none"
  )
```

Figure 2:



Answer 3

For the third visualization, the province names were adjusted for clarity. Additionally, the

provinces of *Yukon* and *Nunavut* were excluded due to missing income data.
The data were then grouped by Canadian regions and income categories. For each subgroup, the number of voters was calculated based on respondents who answered *Yes* to the *p_voted* question, providing the turnout count.

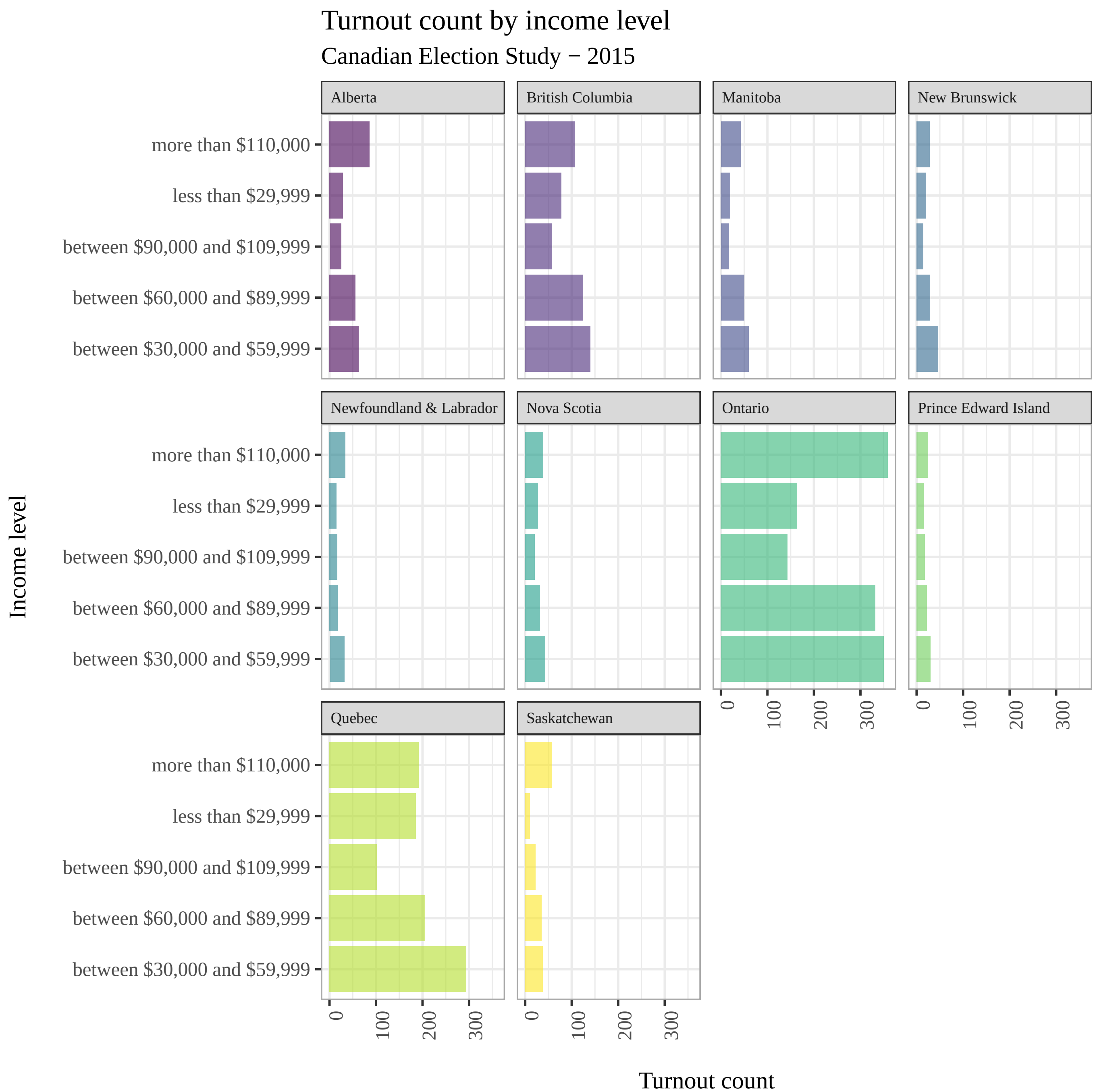
```
ces2015 <- ces2015 |>
  #removing wrongly coded observation
  filter(province != 1000) |>
  mutate(
    province_long = recode(
      province,
      "bc" = "British Columbia",
      "Alberta" = "Alberta",
      "Sask" = "Saskatchewan",
      "Manitoba" = "Manitoba",
      "Ontario" = "Ontario",
      "Quebec" = "Quebec",
      "nb" = "New Brunswick",
      "ns" = "Nova Scotia",
      "pei" = "Prince Edward Island",
      "Nfld" = "Newfoundland & Labrador",
      "nwt" = "Northwest Territories",
      "Yukon" = "Yukon",
      "nunavut" = "Nunavut"
    )
  ) |>
  filter(!province_long %in% c("Yukon", "Nunavut"))

ces2015_viz3 <- ces2015 |>
  #removing non-income level valuesp
  filter(!income_full %in% c("", ".d", ".r")) |>
  group_by(province_long, income_full) |>
  summarise(
    yes_count = sum(p_voted == "Yes", na.rm = TRUE),
    .groups = "drop"
  )
```

Figure 3 reports the histograms counts of turnout, hence the count of how many stated to have voted, by Canadian region and income level. The regions of **Ontario** and **Quebec** showcase the highest overall turnout count.

```
ggplot(data = ces2015_viz3, aes(x = income_full, y = yes_count, fill =
    province_long)) +
  geom_col(alpha = 0.6) +
  scale_fill_viridis_d(name = "province_long") +
  facet_wrap(vars(province_long)) +
  coord_flip()+
  labs(title = "Turnout count by income level",
```

```
      subtitle = "Canadian Election Study - 2015",
      x = "Income level\n",
      y = "\nTurnout count") +
  theme_bw() +
  theme(
    text = element_text(family = "Times"),
    strip.text = element_text(size = 7 , hjust = 0),
    panel.border = element_rect(color = "darkgrey"),
    axis.text.x = element_text(angle = 90, hjust = 1),
    legend.position = "none"
  )
```

Figure 3:



Answer 4

A personalised theme, **PS03_theme**, is created. The theme has all text set in Times New

Roman, with bold plot titles in black, subtitles in dark grey, and captions in lighter grey italics. Facet labels are bold with a dark grey background. The plot background is a cream color, and major grid lines are light grey. Axis titles are bold, while x-axis labels are rotated 90° for readability. Panel borders are dark grey, and the legend is removed for simplicity.

```
PS03_theme <- theme_bw(base_family = "Times", base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = rel(1.5), hjust = 0.5),
    plot.subtitle = element_text(face = "plain", size = rel(1.2), color = "grey20", hjust = 0.5),
    plot.caption = element_text(face = "italic", size = rel(0.8), color = "grey30", hjust = 0.5),
    strip.text = element_text(face = "bold", size = 9, hjust = 0.5),
    plot.title.position = "plot",
    plot.caption.position = "plot",
    strip.background = element_rect(fill = "grey90", color = NA),
    axis.title = element_text(face = "bold", hjust = 0.5),
    axis.text.x = element_text(angle = 90, hjust = 1),
    panel.border = element_rect(color = "darkgrey", fill = NA),
    panel.background = element_rect(fill = "#FFF8E7", color = NA),
    panel.grid.major = element_line(color = "grey85"),
    legend.position = "none"
  )
```

The key pattern highlighted in the figure is the **U-shape of the histograms**. The lowest turnout counts are observed in the second or third income category across regions. To verify this pattern, the income category with the lowest turnout in each region was identified and stored in the variable **low_count**. Notably, both the lowest and the highest income levels consistently exhibit higher turnout counts compared to the middle-income categories.

```
low_count <- ces2015_viz3 |>
  group_by(province_long)|>
  slice_min(yes_count, n = 1) |>
  ungroup()
```

Figure 4 reports *Figure 3*, but formatted with the **PS03_theme**. A title indicating the u-shaped pattern is added, as well as an informative sub-title and caption. Annotation reporting the lowest turnout count is added.

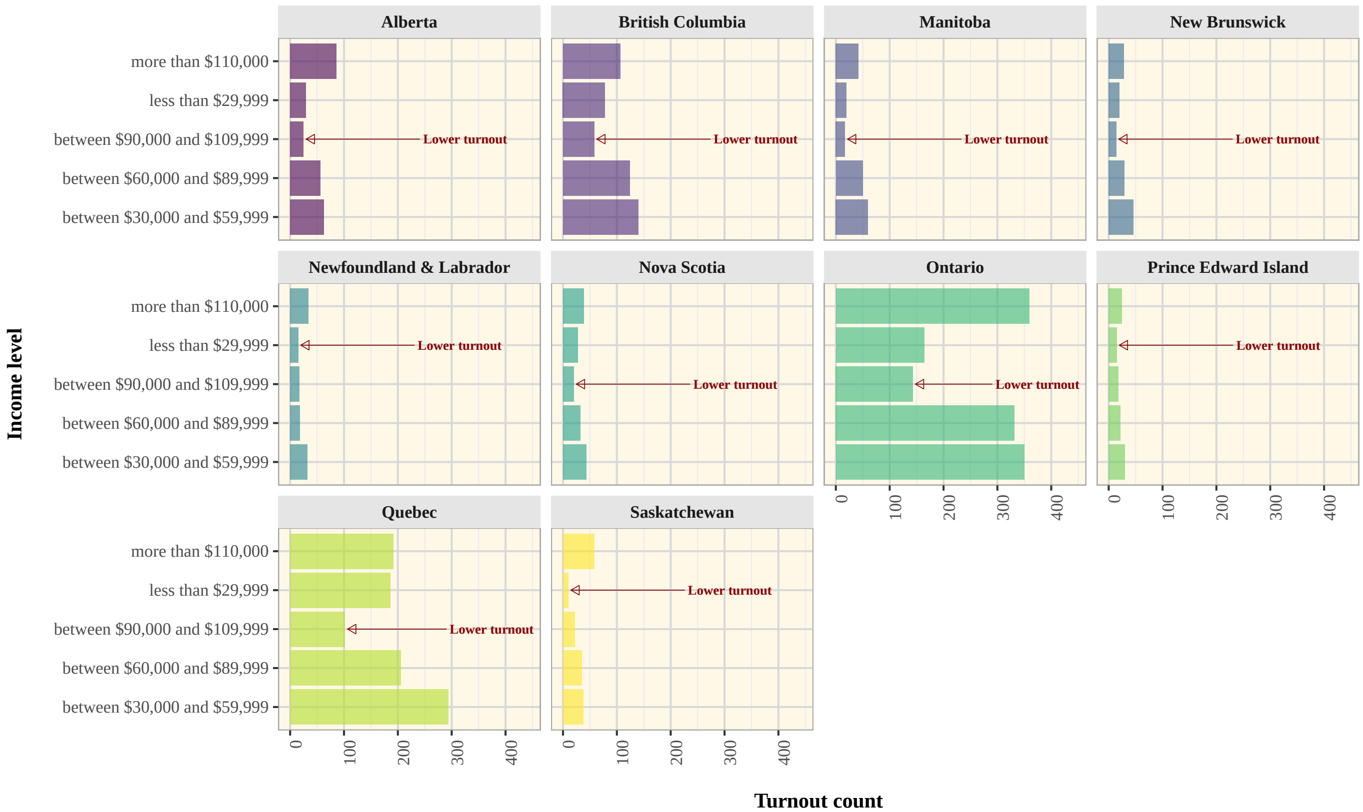
```
ggplot(data = ces2015_viz3, aes(x = income_full, y = yes_count, fill =
    province_long)) +
  geom_col(alpha = 0.6) +
  scale_fill_viridis_d(name = "province_long") +
  facet_wrap(vars(province_long)) +
  coord_flip()+
  geom_text_repel(data = low_count,
```

```
                    aes(x = income_full, y = yes_count, label = "Lower turnout"),
                    nudge_y = 300,
                    size = 2.5,
                    color = "darkred",
                    fontface = "bold",
                    family = "Times",
                    arrow = arrow(length = unit(0.07, "in"), type = "closed"),
                      segment.color = "darkred",
                      segment.size = 0.2)+
  labs(
    title = "Turnout count shows a U-shaped pattern across income levels in each region",
    subtitle = "Number of people who voted in each income bracket, for each of the Canadian regions (2015 CES)",
    caption = "Data source: Canadian Election Study 2015. \nCounts are unweighted. Only respondents reporting Yes for voting were included in the count. Only respondents reporting income were kept.
    The provinces of Yukon and Nunavut were removed for a lack of data.",
    x = "Income level\n",
    y = "\nTurnout count") +
  PS03_theme
```

Figure 4:

Turnout count shows a U−shaped pattern across income levels in each region

Number of people who voted in each income bracket, for each of the Canadian regions (2015 CES)



Data source: Canadian Election Study 2015.
Counts are unweighted. Only respondents reporting Yes for voting were included in the count. Only respondents reporting income were kept.
The provinces of Yukon and Nunavut were removed for a lack of data.